

DEVAGYA RAMAN

RESEARCH SCHOLAR (CSIR - CBRI)

Academy of Scientific and Innovative Research at Central Building Research Institute

+91-8218283390
Devagyaraman2010@gmail.com
Haridwar, UK, INDIA
www.linkedin.com/in/devagya-raman

PROFESSIONAL OVERVIEW

A dedicated Geotechnical Engineer with a strong academic and professional background in Mechanically Stabilized Earth (MSE) walls, currently pursuing a Ph.D. on 3D numerical modeling of MSE walls. Extensive experience in geotechnical investigations, including work on the NHRCL Project (Nadiad lab), 400+ geotechnical reports, MSE wall, drainage, and rotary design for the AAI Sikkim RESA project, soil erosion mitigation for ordnance factories (CFEES-DRDO), PGA analysis for the Omkareshwar 108 ft Shankaracharya statue, and SPT Vs correlation for Noida city.

PROFESSIONAL EXPERIENCE

PROJECT ASSOCIATE 1 CSIR-CBRI

Airport Authority of India ,Sikkim

RESA development Project

- **Designed Mechanically Stabilized Earth (MSE) Walls** for both single-tier and multi-tier structures.
- **Multi-tier walls** include **26 cross sections**, while **single-tier walls** consist of **15 cross sections**.
- **Wall height range: 50m to 103m** (multi-tier) and **3m to 14.4m** (single-tier).
- **Performed numerical modeling** using **Slide v6** and drafting with **AutoCAD 2022**.
- **Calculated stresses for the Crash Fire Truck (CFT)** and designed the rotary as per **CFT and drainage requirements**.

108 ft Statue of Oneness and Other Adjoining Structures in Omkareshwar, Madhya Pradesh

Ground Response Analysis, PGA estimation using MASW Scan Line

- **Performed Ground Response Analysis** to assess site-specific seismic conditions.
- **Evaluated Peak Ground Acceleration (PGA)** for structural stability and seismic design considerations.
- **Conducted Geotechnical Investigations** to analyze soil properties and site behavior.
- **Prepared Detailed Reports** based on seismic and geotechnical assessments for structural safety compliance.

Soil erosion mitigation for the traverses of the Ordnance Factory (ODF)

Centre for Fire, Explosive and Environment Safety (CFEES) ,Project Of DRDO

- **Conducted geotechnical investigations** and **prepared detailed reports** for soil erosion mitigation across **seven sites** in India, including **Badmal ODF, Khamaria ODF, Bhandara ODF, Jaisalmer Army & Air Force Station, Digarua Air Force Station, Pune ODF & HEMRL, and Kashmir Kundru Army Station..**
- **Performed numerical modeling** to assess erosion patterns and optimize mitigation strategies..

EDUCATION

M.tech

(Soil Mechanics and Foundation Design) 7.75 CGPA, 69.75%

National Institute of Technology Kurukshetra ,2021

Bachelor's degree

(Civil Engineering) 66.12%

College of Engineering Roorkee, Roorkee 2017

Senior Secondary School

St. Mary's Sr. Sec School Jwalapur Haridwar

High School

Sapience School Vikasnagar Dehradun 8.8 CGPA 83.6%

SKILLS

- Communication
- Teamwork
- Critical thinking
- Time management
- Leadership
- Adaptability

ACHIEVEMENTS

2nd prize Poster presentation

EGCON Kolkata International Conference

ISEG , 2022

CFEES DRDO Project continued...

- **Performed a literature review** comparing **CO₂ emissions** of conventional techniques with sustainable alternatives.
- **Developed site-specific mitigation strategies** based on **soil properties, climate, and rainfall variations** using **IRC, MORTH, and RDSO guidelines**.
- **Estimated costs** based on **geosynthetics and natural materials** for effective erosion control.
- **Conducted a market survey** to evaluate available **products and companies** in the sector.

DRDO Project ,Landslide Hazard Mapping of North East division

- Digitized 3,000 landslides and Land Use Land Cover (LULC) features
- **Integrated geospatial data** for effective hazard mapping and mitigation planning

GEOTECHNICAL ENGINEER IGS

National High-Speed Rail Corporation Limited, Mumbai-Ahmedabad Bullet Train Project

- **Geotechnical Report Analysis:** Reviewed and interpreted **hundreds of SPT and UDS reports** to assess soil properties.
- **Soil Classification & Strength Assessment**
- **Quality Control & Data Management.**

SEMINARS

- **Advances in Geosynthetics Presented to Dr VK Arora**
[Soil reinforcement and advances with geosynthetics | PPT](#)
- **Testing Of Geosynthetics Presented to Dr Ashwini Jain**
[GEOSYNTHETIC TESTING TECHNIQUES | PPT](#)

M.TECH THEISIS

- Strength improvement of expansive soil treated with Jute Fiber Reinforcement

RESEARCH PAPERS

Journal

S.no	Journal	Paper	Year
1	Journal of Engineering geology	Optimizing the Strength of Expansive Soil with Integration of Jute Fiber Reinforcement	2024
2	Scientific Reports Nature	Seismic site characterization and development of SPT(N) to shear wave velocity correlation of Noida city	2025

International Conference

S.no	Conference	Paper	Year
1	6th World Congress on Disaster Management (WCDDM 2023)	Ground Response Analysis for the Museum Hall Complex and Other Adjoining Structures at the Statue of Oneness Site in Omkareshwar, Madhya Pradesh”	2023

PROFESSIONAL DEVELOPMENT

Plaxis 2D/3D 2 weeks workshop

PEC Chandigarh

2024

Advanced SAR Data Processing for Ground Surface Movement'

IIRS Dehradun

2024

SOFTWARES

Plaxis 2D/3D

Slide 2d

Autocadd

GIS

Primavera

Stadd Pro

SPORTS

- **Coordinator Junior national rowing championship**
- **National Player:** Captain of team Uttarakhand and presented in Senior National Rowing Championship and also participated in All INDIA INTERUNIVERSITY ROWING CHAMPIONSHIP both event held at Hyderabad hussain sagar lake

EXTRA ACTIVITIES

- **Eco Club member,2013-2017**
- **PETA WORKSHOP**
- **HARITIMA FEST**
- **COORDINATED MARATHON in Haritima Fest**

REFERENCES

Dr Anindya Pain , Principal Scientist ,GEGH , CSIR CBRI (INAE YOUNG ASSOCIATE 2024)
pain_anindya@cbri.res.in

Dr. Ashwani Jain , Department of Civil Engineering National Institute of Technology, Kurukshetra-136119(India)
ashwani.jain@nitkkr.ac.in